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Semiconductor device and data storage system including the same

Abstract

A semiconductor device and a data storage system including the same are provided. The semiconductor device including a plate layer, a pattern structure on the plate layer, an upper pattern layer on the pattern structure, an upper structure including a stack structure and a capping insulating structure covering at least a portion of the stack structure, the stack structure including interlayer insulating layers and gate layers alternately stacked on each other, and separation structures and vertical memory structures penetrating through the upper structure, the upper pattern layer, and the pattern structure, and extending into the plate layer may be provided.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims benefit of priority to Korean Patent Application No. 10-2021-0075945 filed on Jun. 11, 2021 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

(2) The present inventive concepts relate to semiconductor devices and/or data storage systems including the same.

(3) In an electronic system requiring data storage, a semiconductor device capable of storing high-capacity data may be required. Accordingly, methods for increasing data storage capacity of semiconductor devices are being studied. For example, as a method for increasing data storage capacity of a semiconductor device, a semiconductor device including memory cells arranged three-dimensionally, instead of memory cells arranged two-dimensionally, has been proposed.

SUMMARY

(4) An aspect of the present inventive concepts is to provide semiconductor devices capable of improving a degree of integration.

(5) An aspect of the present inventive concepts is to provide data storage systems including a semiconductor device capable of improving a degree of integration.

(6) A semiconductor device according to an aspect of the present inventive concepts may include a plate layer, a pattern structure on the plate layer, an upper pattern layer on the pattern structure, an upper structure including a stack structure being on the upper pattern layer and including interlayer insulating layers and gate layers alternately stacked each other, a capping insulating structure covering at least a portion of the stack structure, separation structures penetrating through the upper structure, the upper pattern layer, and the pattern structure, and extending into the plate layer, and vertical memory structures penetrating through the upper structure, the upper pattern layer, and the pattern structure between the separation structures, extending into the plate layer, and contacting the plate layer, wherein each of the separation structures includes a lower portion disposed in the plate layer and including a first lower portion contacting the plate layer, a second lower portion penetrating through the pattern structure, and a third lower portion penetrating through the upper pattern layer, and an upper portion penetrating through the upper structure on the lower portion, and a maximum width of the first lower portion is different from a maximum width of the third lower portion.

(7) A semiconductor device according to an aspect of the present inventive concepts includes a plate layer, an upper structure including a stack structure and a capping insulating structure, the stack structure on the plate layer and including interlayer insulating layers and gate layers alternately stacked on each other, the capping insulating structure covering at least a portion of the stack structure, separation structures penetrating through at least a portion of the upper structure, and vertical memory structures penetrating through the upper structure between the separation structures, wherein a first separation structure among the separation structures comprises a first line

portion extending in a first direction, parallel to an upper surface of the plate layer, and a plurality of first pillar portions being on a different height level from the first line portion and spaced apart from each other in the first direction, and the plurality of first pillar portions are at a level lower than that of an uppermost gate layer among the gate layers, and at a level higher than that of a lowermost gate layer among the gate layers.
(8) A data storage system according to an aspect of the present inventive concepts includes a semiconductor device including an input/output pad, and a controller electrically connected to the semiconductor device through the input/output pad and configured to control the semiconductor device, wherein the semiconductor device includes a plate layer, a pattern structure on the plate layer, an upper pattern layer on the pattern structure, an upper structure including a stack structure and a capping insulating structure covering at least a portion of the stack structure, the stack structure on the plate layer and including interlayer insulating layers and gate layers alternately stacked on each other, separation structures penetrating through the upper structure, the upper pattern layer, and the pattern structure, and extending into the plate layer, and vertical memory structures penetrating through the upper structure, the upper pattern layer, and the pattern structure between the separation structures, extending into the plate layer, and contacting the plate layer, wherein each of the separation structures includes a lower portion disposed in the plate layer and including a first lower portion contacting the plate layer, a second lower portion penetrating through the pattern structure, and a third lower portion penetrating through the upper pattern layer, and an upper portion penetrating through the upper structure on the lower portion, and a maximum width of the first lower portion is different from a maximum width of the third lower portion.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) The above and other aspects, features, and advantages of the present inventive concepts will be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:
(2) FIGS. **1**, **2**A, **2**B, **3** and **4** are views illustrating a semiconductor device according to an example embodiment of the present inventive concepts.
(3) FIG. **5** is a cross-sectional view illustrating a modified example of a semiconductor device according to an example embodiment of the present inventive concepts.
(4) FIG. **6** is a cross-sectional view illustrating a modified example of a semiconductor device according to an example embodiment of the present inventive concepts.
(5) FIGS. **7** and **8** are views illustrating a modified example of a semiconductor device according to an example embodiment of the present inventive concepts.
(6) FIGS. **9**A, **9**B, **9**C, **9**D, **9**E, **10**A, **10**B, and **11** are views illustrating a modified example of a semiconductor device according to an example embodiment of the present inventive concepts.
(7) FIGS. **12**A and **12**B are views illustrating a modified example of a semiconductor device according to an example embodiment of the present inventive concepts.
(8) FIGS. **13** and **14** are views illustrating a modified example of a semiconductor device according to an example embodiment of the present inventive concepts.
(9) FIG. **15** is a cross-sectional view illustrating a modified example of a semiconductor device according to an example embodiment of the present inventive concepts.
(10) FIGS. **16**A, **16**B, and **16**C are plan views illustrating a modified example of a semiconductor device according to an example embodiment of the present inventive concepts.
(11) FIGS. **17**A and **17**B are views illustrating a modified example of a semiconductor device according to an example embodiment of the present inventive concepts.
(12) FIGS. **18** to **25**F are views illustrating an example of a method of forming a semiconductor

device according to an example embodiment of the present inventive concepts.
(13) FIG. **26** is a cross-sectional view illustrating a modified example of a semiconductor device according to an example embodiment of the present inventive concepts.
(14) FIG. **27** is a cross-sectional view illustrating a modified example of a semiconductor device according to an example embodiment of the present inventive concepts.
(15) FIG. **28** is a view schematically illustrating a data storage system including a semiconductor device according to an example embodiment of the present inventive concept.
(16) FIG. **29** is a perspective view schematically illustrating a data storage system including a semiconductor device according to an example embodiment of the present inventive concepts.

DETAILED DESCRIPTION

(17) Hereinafter, terms such as “upper,” “intermediate,” and “lower” may be replaced with other terms, for example, “first,” “second,” and “third,” to describe the elements of the specification. Terms such as “first,” “second,” and “third” may be used to describe various components, but the components are not limited by the terms. A “first component” may be called a “second component,” or may be named as another term, distinguishable from other components.
(18) When the terms “about” or “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value includes a manufacturing or operational tolerance (e.g., ±10%) around the stated numerical value. Moreover, when the words “generally” and “substantially” are used in connection with geometric shapes, it is intended that precision of the geometric shape is not required but that latitude for the shape is within the scope of the disclosure. Further, regardless of whether numerical values or shapes are modified as “about” or “substantially,” it will be understood that these values and shapes should be construed as including a manufacturing or operational tolerance (e.g., ±10%) around the stated numerical values or shapes.
(19) First, a semiconductor device according to an example embodiment of the present inventive concepts will be described with reference to FIGS. **1**, **2**A, and **2**B. FIG. **1** may be a plan view schematically illustrating a semiconductor device according to an example embodiment of the present inventive concepts, FIG. **2**A may be a cross-sectional view schematically illustrating a region taken along line I-I′ and a region taken along line II-II′ of FIG. **1**, and FIG. **2**B may be a cross-sectional view schematically illustrating a region taken along line III-III′ and a region taken along line IV-IV′ of FIG. **1**.
(20) Referring to FIGS. **1**, **2**A, and **2**B, a semiconductor device **1** according to an example embodiment may include a peripheral circuit structure **3**, a plate layer **15**, an upper structure **59**, vertical memory structures **65**, vertical support structures **75**, and separation structures **89**.
(21) The peripheral circuit structure **3** may include a semiconductor substrate **5**, an device isolation layer **7***s* defining an active region **7***a* in the semiconductor substrate **5**, a peripheral circuit **9** on the semiconductor substrate **5**, a circuit interconnection **11** electrically connected to the peripheral circuit **9** on the semiconductor substrate **5**, and a lower insulating layer **13** covering the peripheral circuit **9** and the circuit interconnection **11** on the semiconductor substrate **5**. The peripheral circuit **9** may include a peripheral gate **9***a* on the active region **7***a*, and a peripheral source/drain **9***b* disposed in the active region **7***a* on both sides of the peripheral gate **9***a*.
(22) The plate layer **15** may be disposed on the peripheral circuit structure **3**. The plate layer **15** may include a silicon layer, for example, a doped silicon layer. For example, the plate layer **15** may include a silicon layer having N-type conductivity.
(23) The semiconductor device **1** may further include a pattern structure **19** on the plate layer, and an upper pattern layer **27** on the pattern structure **19**. The upper pattern layer **27** may include a contact portion **27***a* penetrating through the pattern structure **19** and contacting the plate layer **15**. The upper pattern layer **27** may include a silicon layer, for example, a doped silicon layer. For example, the upper pattern layer **27** may be formed of a silicon layer having N-type conductivity.
(24) The pattern structure **19** may include a first pattern layer **19***a* and a second pattern layer **19***b*,

separated by the contact portion **27***a* of the upper pattern layer **27**. The first pattern layer **19***a* and the second pattern layer **19***b* may be disposed on the same height level.
(25) The first pattern layer **19***a* may include a silicon layer, for example, a doped silicon layer. For example, the first pattern layer **19***a* may be formed of a silicon layer having N-type conductivity. The second pattern layer **19***b* may include at least two layers, for example, a first intermediate layer **21***a*, a second intermediate layer **21***b*, and a third intermediate layer **21***c*, sequentially stacked. The first and third intermediate layers **21***a* and **21***c* may be formed of silicon oxide, and the second intermediate layer **21***b* may be formed of silicon nitride.
(26) The upper structure **59** may include a stack structure ST and a capping insulating structure **53**.
(27) The stack structure ST may include interlayer insulating layers **30** and **45** and gate layers **33***g* and **48***g*, alternately stacked on the upper pattern layer **27**.
(28) The interlayer insulating layers **30** and **45** may be lower interlayer insulating layers **30** spaced apart from each other in a vertical (Z) direction, perpendicular to an upper surface of the plate layer **15**, and upper interlayer insulating layers **45** spaced apart from each other in the vertical (Z) direction on a level higher than that of the lower interlayer insulating layers **30**.
(29) The gate layers **33***g* and **48***g* include lower gate layers **33***g* spaced apart from each other in the vertical (Z) direction, and upper gate layers **48***g* spaced apart from each other in the vertical (Z) direction on a level higher than the lower gate layers **33***g*.
(30) The lower interlayer insulating layers **30** and the lower gate layers **33***g* may be alternately and repeatedly stacked to form a lower stack structure ST_L. The upper interlayer insulating layers **45** and the upper gate layers **48***g* may be alternately and repeatedly stacked to form an upper stack structure ST_U. The stack structure ST may include the lower stack structure ST_L and the upper stack structure ST_U on the lower stack structure ST_L.
(31) An uppermost layer among the lower interlayer insulating layers **30** and the lower gate layers **33***g* may be an uppermost lower interlayer insulating layer **30**U.
(32) Among the upper interlayer insulating layers **45** and the upper gate layers **48***g*, an uppermost layer may be an uppermost upper interlayer insulating layer **45**U, and a lowermost layer may be a lowermost upper interlayer insulating layer **45**L.
(33) At least one side of the stack structure ST may have a stepped shape. For example, the stack structure ST may have a substantially flat upper surface in a first region MCA on the plate layer **15** and may have a stepped shape in a second region SA on the plate layer **15**.
(34) The first region MCA may be a memory cell array region, and the second region SA may be a stepped region or a gate contact region. Hereinafter, the first region MCA will be referred to as a ‘memory cell array region’ and the second region SA will be referred to as a ‘stepped region.’
(35) In the stepped region SA, the semiconductor device **1** may further include gate contact plugs **92** electrically connected to the gate layers **33***g* and **48***g*.
(36) The capping insulating structure **53** may include a lower capping insulating layer **36** covering at least a portion of the lower stack structure ST_L having a stepped shape, and an upper capping insulating layer **51** covering at least a portion of the upper stack structure ST_U having a stepped shape on the lower capping insulating layer **36**.
(37) The separation structures **89** may penetrate through the upper structure **59**, the upper pattern layer **27**, and the pattern structure **19**, and may extend into the plate layer **15**.
(38) The separation structures **89** may include first separation structures **891** parallel to each other, and a second separation structure **89_2** and a third separation structure **89_3** that are arranged between the first separation structures **89_1** and have opposite end portions, respectively. In other words, a second separation structure **89_2** and a third separation structure **893** may be provided in a row between the first separation structures **89_1**.
(39) The separation structures **89** may further include a connection portion **89***i* connecting a lower portion of the second separation structure **89_2** and a lower portion of the third separation structure **89_3**.

(40) The stack structure ST may further include a stacked connection portion STi disposed between the second separation structure **89_2** and the third separation structure **89_3** and disposed on the connection portion **89***i*.
(41) A portion of the upper pattern layer **27** may be disposed between the stacked connection portion STi and the connection portion **89***i*.
(42) The separation structures **89** may be formed of an insulating material such as silicon oxide, but example embodiments thereof are not limited thereto. For example, each of the separation structures **89** may include a conductive pattern and an insulating spacer covering a side surface of the conductive pattern.
(43) The vertical memory structures **65** may be disposed in the memory cell array region MCA. The vertical memory structures **65** may penetrate through the upper structure **59**, the upper pattern layer **27**, and the first pattern layer **19***a* of the pattern structure **19** between the separation structures **89**, and may extend into the plate layer **15** to contact the plate layer **15**.
(44) The vertical support structures **75** may be disposed in the stepped region SA. The vertical support structures **75** may penetrate through the upper structure **59**, the upper pattern layer **27**, and the first pattern layer **19***a* of the pattern structure **19** between the separation structures **89**, and may extend into the plate layer **15** to contact the plate layer **15**. The vertical support structures **75** may serve as supports for preventing defects such as deformation or collapse of the stack structure ST in the stepped region SA.
(45) Next, a cross-sectional structure of any one of the vertical memory structures **65** and a cross-sectional structure of the stack structure ST will be described with reference to FIG. **3**. FIG. **3** may be a partially enlarged view of portion 'A' of FIG. **2**A.
(46) Referring to FIG. **3** together with FIGS. **1** to **2**B, the vertical memory structure **65** may include an insulating core pattern **71**, a channel layer **69** covering side and bottom surfaces of the insulating core pattern **71**, a pad pattern **73** disposed on the insulating core pattern **71** and contacting the channel layer **69**, and a data storage structure **67** covering at least an outer side surface of the channel layer **69**. The data storage structure **67** may include a first dielectric layer **67***a*, a second dielectric layer **67***b*, and a data storage layer **67***d* between the first and second dielectric layers **67***a* and **67***b*. The second dielectric layer **67***b* may be interposed between the data storage layer **67***d* and the channel layer **69**.
(47) The insulating core pattern **71** may include silicon oxide, for example, silicon oxide formed by an atomic layer deposition process, or silicon oxide having voids formed therein. The second dielectric layer **67***b* may include silicon oxide or silicon oxide doped with impurities. The first dielectric layer **67***a* may include at least one of silicon oxide and a high-k dielectric. The data storage layer **67***d* may include a material capable of storing information by trapping a charge, for example, silicon nitride. The data storage layer **67***d* may include regions capable of storing information in a semiconductor device such as a flash memory device. The channel layer **69** may include a silicon layer, for example, an undoped silicon layer. The pad pattern **73** may include at least one of doped polysilicon, a metal nitride (e.g., TiN), a metal (e.g., W), or a metal-semiconductor compound (e.g., TiSi).
(48) The first pattern layer **19***a* may penetrate through the data storage structure **67**, and may be in contact with the channel layer **69**, and the data storage structure **67** may be separated as an upper portion and a lower portion by the first pattern layer **19***a*. The first pattern layer **19***a* contacting the channel layer **69** may be formed of a silicon layer having N-type conductivity.
(49) On a height level between an uppermost lower gate layer of the lower gate layers **33***g* and a lowermost upper gate layer of the upper gate layers **48***g*, a side surface of the vertical memory structure **65** may include a side slope change portion **65**V.
(50) In the side surface of the vertical memory structure **65**, the side slope change portion **65**V may refer to a portion in which a slope changes between an adjacent upper side surface **65**S**2** and an adjacent lower side surface **65**S**1**. For example, in the side surface of the vertical memory structure

65, the side slope change portion **65**V may refer to a portion of a side surface having a gentle slope between the upper side surface **65**S**2** having a steep slope and the lower side surface **65**S**1** having a steep slope.
(51) The lower and upper gate layers **33***g* and **48***g* may include at least one of doped polysilicon, a metal-semiconductor compound (e.g., TiSi, TaSi, CoSi, NiSi, or WSi), metal nitride (e.g., TiN, TaN, or WN), or a metal (e.g., Ti or W).
(52) The stack structure ST may further include a dielectric layer **50** covering upper and lower surfaces of each of the lower and upper gate layers **33***g* and **48***g*, and extending between each of the lower and upper gate layers **33***g* and **48***g* and the vertical memory structure **65**. The dielectric layer **50** may include a high-k dielectric such as AlO or the like.
(53) In an example, the vertical memory structure **65** and the vertical support structure **75** in FIGS. **1** and **2**A may be simultaneously formed, and may include layers of the same material. For example, the vertical support structure **75** may include layers corresponding to the data storage structure **67**, the channel layer **69**, and the pad pattern **73** of the vertical memory structure **65**, respectively.
(54) Next, a cross-sectional structure of any one of the separation structures **89** will be described with reference to FIG. **4**. FIG. **4** may be a partially enlarged view of portion 'B' of FIG. **2**A.
(55) Referring to FIG. **4** together with FIGS. **1** to **3**, each of the separation structures **89** may include a lower portion **89**_L and an upper portion **89**_U on the lower portion **89**_L.
(56) The lower portion **89**_L may be disposed in a trench **17** of the plate layer **15**, and may include a first lower portion **89**_L**1** contacting the plate layer **15**, a second lower portion **89**_L**2** penetrating through the pattern structure **19**, and a third lower portion **89**_L**3** penetrating through the upper pattern layer **27**. The upper portion **89**_U may be disposed on the lower portion **89**_L, and may penetrate through the upper structure **59**. The upper portion **89**_U may include a first upper portion **89**_U**1** at least penetrating through the first stack structure ST_L, and a second upper portion **89**_U**2** disposed on the first upper portion **89**_U**1** and at least penetrating through the second stack structure ST_U.
(57) A maximum width of the first lower portion **89**_L**1** may be different from a maximum width of the third lower portion **89**_L**3**. For example, a maximum width of the first lower portion **89**_L**1** may be wider than a maximum width of the third lower portion **89**_L**3**.
(58) A maximum width of the second lower portion **89**_L**2** may be different from a maximum width of the third lower portion **89**_L**3**. For example, a maximum width of the second lower portion **89**_L**2** may be wider than a maximum width of the third lower portion **89**_L**3**.
(59) A maximum width of the second lower portion **89**_L**2** may be different from a maximum width of the first lower portion **89**_L**1**. For example, a maximum width of the second lower portion **89**_L**2** may be wider than a maximum width of the first lower portion **89**_L**1**.
(60) A width of the upper portion **89**_U adjacent to the lower portion **89**_L may be narrower than a width of the first lower portion **89**_L**1**.
(61) A width of the upper portion **89**_U adjacent to the lower portion **89**_L may be narrower than a width of the second lower portion **89**_L**2**.
(62) A width of the upper portion **89**_U adjacent to the lower portion **89**_L may be the same as or substantially similar to a width of the third lower portion **89**_L**3**.
(63) In each of the separation structures **89**, a side surface **89**S of the upper portion **89**_U may include a slope change portion **89**V in which a side slope changes, on a height level between an uppermost lower gate layer of the lower gate layers **33***g* and a lowermost upper gate layer of the upper gate layers **48***g*. For example, in each of the separation structures **89**, the side surface **89**S of the upper portion **89**_U may include a first side surface **89**S**1** of the first upper portion **89**_U**1**, a second side surface **89**S**2** of the second upper portion **89**_U**2**, and a slope change portion **89**V having a slope, different from a slope of the first side surface **89**S**1** and a slope of the second side surface **89**S**2**.

(64) When viewed in plan view, in each of the separation structures **89**, a side shape of the first lower portion **89**_L**1** may be different from a side shape of the upper portion **89**_U. For example, in plan view, a side surface of the first lower portion **89**_L**1** may have a straight linear shape, and a side surface of the upper portion **89**_U may have a wavy shape. Because a sidewall of the first lower portion **89**_L**1** may be the same as or substantially similar to a sidewall of the trench **17**, the sidewall of the trench **17** may have a straight linear shape, in plan view.
(65) When viewed in plan view, in each of the separation structures **89**, a side surface of the third lower portion **89**_L**3** may have a wavy shape.
(66) When viewed in plan view, in each of the separation structures **89**, a side surface of the second lower portion **89**_L**2** may have a wavy shape.
(67) Hereinafter, various modifications of a semiconductor device **1** according to an example embodiment of the present inventive concepts will be described. Hereinafter, in describing various modified examples of the above-described semiconductor device **1**, the above-described components of the semiconductor device **1** that may be modified or replaced will be mainly described, and descriptions of the substantially same components as the above-described components, components that may be easily understood from the above-described components, or components that may be easily understood from the drawings described above will be omitted.
(68) First, a modified example of a semiconductor device according to an example embodiment of the present inventive concepts will be described with reference to FIG. **5**. FIG. **5** may be a cross-sectional view schematically illustrating a region taken along line I-I′ and a region taken along line II-II′ of FIG. **1**, to illustrate a modified example of a semiconductor device according to an example embodiment of the present inventive concepts.
(69) Referring to FIG. **5**, a buffer pattern layer **26** may be disposed between a first pattern layer **19***a* and a second pattern layer **19***b* of a pattern structure **19** that are spaced apart from each other. The buffer pattern layer **26** may be formed as a silicon layer. Therefore, the first pattern layer **19***a* and the second pattern layer **19***b* of the pattern structure **19** may be separated by the buffer pattern layer **26**. An upper pattern layer **27** may cover the pattern structure **19** and the buffer pattern layer **26**.
(70) Next, a modified example of a semiconductor device according to an example embodiment of the present inventive concepts will be described with reference to FIG. **6**. FIG. **6** may be a cross-sectional view schematically illustrating a region taken along line I-I′ and a region taken along line II-II′ of FIG. **1**, to illustrate a modified example of a semiconductor device according to an example embodiment of the present inventive concepts.
(71) Referring to FIG. **6**, a vertical support structure **75**′ in the modified example may include a material, different from that of a vertical memory structure **65**. For example, the vertical support structure **75**′ may be formed of an insulating material such as silicon oxide, and may not include a data storage structure (**67** of FIG. **3**), a channel layer (**69** of FIG. **3**), and a pad pattern (**73** of FIG. **3**) of a vertical memory structure (**65** of FIG. **3**).
(72) Next, a modified example of a semiconductor device according to an example embodiment of the present inventive concepts will be described with reference to FIGS. **7** and **8**. FIG. **7** may be a plan view illustrating a modified example of the trench **17** of FIG. **1**, and FIG. **8** may be a cross-sectional view schematically illustrating a region taken along line IIIa-IIIa′ of FIG. **7**, to illustrate a modified example of a semiconductor device according to an example embodiment of the present inventive concepts.
(73) Referring to FIGS. **7** and **8**, in the modified example, a trench **17** may include first trenches **171** overlapping first separation structures **891**, a second trench **17_2** overlapping a second separation structures **89_2**, and a third trench **17_3** overlapping a third separation structure **89_3**.
(74) The second trench **17_2** and the third trench **17_3** may be spaced apart from each other. A plate layer **15** may include a portion **15***p* separating the second trench **17_2** and the third trench **17_3**. The second trench **17_2** and the third trench **17_3** may be spaced apart from each other.
(75) The connection portion **89***i* of FIG. **2**B, described above, may be omitted, and a second

separation structure **89_2** and a third separation structure **89_3** may be spaced apart from each other.
(76) An upper pattern layer **27** may include a portion **27***a*′ contacting the portion **15***p* of the plate layer **15** separating the second trench **17_2** and the third trench **17_3** from each other.
(77) The second separation structure **89_2** and the third separation structure **89_3** may be separated from each other by the portion **15***p* of the plate layer **15**, the portion **27***a*′ of the upper pattern layer **27**, and a stacked connection portion STi, as described in FIG. **2**B.
(78) Next, a modified example of a semiconductor device according to an example embodiment of the present inventive concepts will be described with reference to FIGS. **9**A to **9**E, **10**A, **10**B, and **11**. In FIGS. **9**A to **11**, FIGS. **9**A to **9**E may be plan views schematically illustrating a planar shape according to a height level in a modified example of a semiconductor device according to an example embodiment of the present inventive concepts, FIG. **10**A may be a cross-sectional view schematically illustrating a region taken along line Ib-Ib′ and a region taken along line IIb-IIb′ in FIGS. **9**A to **9**E, FIG. **10**B may be a cross-sectional view schematically illustrating a region taken along line IIIb-IIIb′ and a region taken along line IVb-IVb′ in FIGS. **9**A to **9**E, and FIG. **11** may be a partial enlarged view schematically illustrating portion 'Ba' of FIG. **10**A.
(79) Referring to FIGS. **9**A to **9**E, **10**A, **10**B, and **11**, a semiconductor device **1***a* in the modified example may include a peripheral circuit structure **3**, a plate layer **15**, a pattern structure **19**, an upper pattern layer **27**, an upper structure **59**, vertical memory structures **65**, and vertical support structures **75**, to be substantially identical to those described above.
(80) The semiconductor device **1***a* may include separation structures **189** that may replace the separation structures **89** described above.
(81) The separation structures **189** may include first separation structures **1891** parallel to each other, and a second separation structure **189_2** and a third separation structure **1893**, arranged between the first separation structures **189_1** and having opposite end portions, respectively. In other words, the second separation structure **189_2** and the third separation structure **1893** may be provided in a row between the first separation structures **189_1**.
(82) The separation structures **189** may further include a connection portion **189***i* connecting a lower portion of the second separation structure **189_2** and a lower portion of the third separation structure **189_3**. The connection portion **189***i* may be the same as or substantially similar to the connection portion **89***i* described with reference to FIG. **2**B.
(83) Each of the separation structures **189** may include a lower portion **189**L and an upper portion **189**U on the lower portion **189**L.
(84) In each of the separation structures **189**, the lower portion **189**L may include a first lower portion **189**L**1** disposed in a trench **17** of the plate layer **15** and contacting the plate layer **15**, a second lower portion **189**L**2** penetrating through the pattern structure **19**, and a plurality of third lower portions **189**L**3** penetrating through the upper pattern layer **27**. The upper portion **189**U may be disposed on the lower portion **189**L, and may penetrate through the upper structure **59**.
(85) Each of the first and second lower portions **189**L**1** and **189**L**2** may have a linear shape extending in a first (X) direction.
(86) The plurality of third lower portions **189**L**3** may have pillar shapes extending in the vertical (Z) direction from the line-shaped second lower portion **189**L**2** and spaced apart from each other in the first (X) direction. The plurality of third lower portions **189**L**3** may be referred to as pillar portions. The second lower portion **189**L**2** may be referred to as a line portion.
(87) The upper portion **189**U may include a first upper portion **189**U**1** at least penetrating through a lower stack structure ST_L, and a second upper portion **189**U**2** at least penetrating through an upper stack structure ST_U.
(88) The first upper portion **189**U**1** may include a plurality of first portions **189**U**1_1** having pillar shapes and spaced apart from each other in the first (X) direction, a second portion **189**U**1_2** disposed on the plurality of first portions **189**U**1_1** and having a linear shape extending in the first

(X) direction, and third portions **189**U**1_3** disposed on the second portion **189**U**1_2**, having a pillar shape, and spaced apart from each other in the first (X) direction.
(89) The plurality of first portions **189**U**1_1** may be referred to as pillar portions, and the third portions **189**U**1_3** may be referred to as pillar portions. The second portion **189**U**1_2** may be referred to as a line portion.
(90) The plurality of first portions, that is the first pillar portions **189**U**1_3** may be disposed on a lever lower than an uppermost gate layer among gate layers **33***g* and **48***g* and on a level higher than that of a lowermost gate layer among gate layers **33***g* and **48***g*.
(91) The plurality of first portions **189**U**1_1** may penetrate through a portion of a lowermost lower interlayer insulating layer **30**L among lower interlayer insulating layers **30**.
(92) The second portion **189**U**1_2** may cover the plurality of first portions **189**U**1_1**, and may extend to penetrate through a portion of the lowermost lower interlayer insulating layer **30**L while penetrating through a portion of an uppermost lower interlayer insulating layer **30**U.
(93) Therefore, the second portion **189**U**1_2** may penetrate through lower gate layers **33***g* and the lower interlayer insulating layers **30** between the lower gate layers **33***g*. The third portions **189**U**1_3** may penetrate through a remaining portion of the uppermost lower interlayer insulating layer **30**U.
(94) The second upper portion **189**U**2** may include a plurality of fourth portions **189**U**2_1** overlapping the third portions **189**U**1_3**, having a pillar shape, and spaced apart from each other in the first (X) direction, a fifth portion **189**U**2_2** disposed on the plurality of fourth portions **189**U**2_1** and having a linear shape extending in the first (X) direction on the portions, and sixth portions **189**U**2_3** disposed on the fifth portion **189**U**2_2**, having a pillar shape, and spaced apart from each other in the first (X) direction.
(95) The plurality of fourth portions **189**U**2_1** may be referred to as pillar portions. The fifth portion **189**U**2_2** may be referred to as a line portion. The sixth portions **189**U**2_3** may be referred to as pillar portions.
(96) The plurality of fourth portions **189**U**2_1** may penetrate through a portion of a lowermost upper interlayer insulating layer **45**L. The fifth portion **189**U**2_2** may cover the plurality of fourth portions **189**U**2_1**, and may extend to penetrate through a portion of the lowermost upper interlayer insulating layer **45**L while penetrating through a portion of an uppermost upper interlayer insulating layer **45**U. Therefore, the fifth portion **189**U**2_2** may penetrate through upper gate layers **48***g* and upper interlayer insulating layers **45** between the upper gate layers **48***g*. The sixth portions **189**U**2_3** may penetrate through a remaining portion of the uppermost upper interlayer insulating layer **45**U.
(97) On the same height level as the plurality of third portions **189**U**1_3**, the uppermost lower interlayer insulating layer **30**U may be connected as a single layer without being separated by the separation structures **189**. On the same height level as the plurality of fourth portions **189**U**2_1**, the lowermost upper interlayer insulating layer **45**L may be connected as a single layer without being separated by the separation structures **189**. On the same height level as the sixth portions **189**U**2_3**, the uppermost upper interlayer insulating layer **45**U may be connected as a single layer without being separated by the separation structures **189**.
(98) The uppermost lower interlayer insulating layer **30**U and the lowermost upper interlayer insulating layer **45**L, located on the same height level as the plurality of third portions **189**U**1_3** and the plurality of fourth portions **189**U**2_1**, and overlapping the second portion **189**U**1_2** and the fifth portion **189**U**2_2** having a linear shape extending in the first (X) direction, may be defined as an intermediate support structure SP**1**. Therefore, at least a portion of the intermediate support structure SP**1** may be disposed on the same level as at least a portion of any one interlayer insulating layer among the interlayer insulating layers.
(99) The uppermost upper interlayer insulating layer **45**U located on the same height level as the sixth portions **189**U**2_3** and overlapping the second portion **189**U**1_2** and the fifth portion

189U**2_2**, extending in the first (X) direction, may be defined as an upper support structure SP**2**.
(100) In some example embodiments, the intermediate and upper support structures SP**1** and SP**2** may serve to mitigate or prevent a stack structure ST from being collapsed or deformed. Therefore, reliability, durability, or productivity of the semiconductor device **1***a* may be improved.
(101) Referring to FIGS. **9**A to **9**E, FIG. **9**A may be a plan view on a height level on which a lowermost lower gate layer **33***g*L among the lower gate layers **33***g* is located, FIG. **9**B may be a plan view on a height level on which a lowermost lower interlayer insulating layer **30**L adjacent to the lowermost lower gate layer **33***g*L is located, FIG. **9**C may be a plan view on a height level on which an uppermost lower interlayer insulating layer **30**U and a lower support structure SP**1** are located, FIG. **9**D may be a plan view on a height level on which an uppermost upper gate layer **48***g*U is located, and FIG. **9**E may be a plan view on a height level on which an uppermost upper interlayer insulating layer **45**U and an upper support structure SP**2** are located.
(102) Referring to FIGS. **9**A to **9**E, the second portion **189**U**1_2** and the fifth portion **189**U**2_2** having a linear shape extending in the first (X) direction may have a side surface of a wave pattern, and portions of the separation structures **189** located on the same height level as the intermediate and upper support structures SP**1** and SP**2**, for example, each of the plurality of third portions **189**U**1_3** and the sixth portions **189**U**2_3** may have a circular shape.
(103) Next, a modified example of a semiconductor device according to an example embodiment of the present inventive concepts will be described with reference to FIGS. **12**A and **12**B. FIG. **12**A may be a cross-sectional view schematically illustrating a region taken along line Ib-Ib′ and a region taken along line IIb-IIb′ in FIGS. **9**A to **9**E, and FIG. **12**B may be a cross-sectional view schematically illustrating a region taken along line IIIb-IIIb′ and a region taken along line IVb-IVb′ in FIGS. **9**A to **9**E.
(104) Referring to FIGS. **12**A and **12**B, the plurality of third lower portions **189**L**3** spaced apart from each other in the first (X) direction and formed in a pillar shape and the plurality of first portions **189**U**1_1**, described in FIGS. **10**A and **11**, may be modified in a linear shape extending in the first (X) direction. Therefore, separation structures **189** may include a lower portion **189**L′ in which the plurality of third lower portions **189**L**3** formed in the pillar shape, as described with reference to FIGS. **10**A and **11**, are modified to have a linear shape, and an upper portion **189**U′ in which the plurality of first portions **189**U**1_1** formed in the pillar shape, as described with reference to FIGS. **10**A and **11**, are modified to have a linear shape.
(105) Next, a modified example of a semiconductor device according to an example embodiment of the present inventive concepts will be described with reference to FIGS. **13** and **14**. FIG. **13** may be a cross-sectional view schematically illustrating a region taken along line Ib-Ib′ and a region taken along line IIb-IIb′ in FIGS. **9**A to **9**E, and FIG. **14** may be a partial enlarged view schematically illustrating portion ‘Bb’ in FIG. **13**.
(106) Referring to FIGS. **13** and **14**, the plate layer **15** having the trench **17**, described in FIG. **10**A, may be modified as a plate layer **15**′, not having the trench **17**. In separation structures **189**, the lower portion **189**L having a lower surface located on a level lower than that of the lower surface of the vertical memory structures **65**, described in FIG. **10**A may be replaced with a lower portion **189**L″ having a lower surface located on a level higher than that of the lower surface of the vertical memory structures **65**.
(107) In the separation structures **189**, the lower portion **189**L″ may include a portion **189**La penetrating through a first pattern layer **19***a*, and a portion **189**Lb penetrating through an upper pattern layer **27**.
(108) In the lower portion **189**L″, both the portion **189**La penetrating through the first pattern layer **19***a* and the portion **189**Lb penetrating through the upper pattern layer **27** may have a width narrower than a width of a second portion **189**U**1_2** of an upper portion **189**U that is adjacent to the lower portion **189**L″.
(109) In the lower portion **189**L″, the portion **189**Lb penetrating through the upper pattern layer **27**

may be formed to have a plurality of pillar shapes spaced apart from each other in the first (X) direction.
(110) Next, a modified example of a semiconductor device according to an example embodiment of the present inventive concepts will be described with reference to FIG. **15**.
(111) FIG. **15** may be a cross-sectional view schematically illustrating a region taken along line Ib-Ib′ and a region taken along line IIb-IIb′ in FIGS. **9**A to **9**E.
(112) Referring to FIG. **15**, the lower portion **189**L″ including the portion formed to have the plurality of pillar shapes spaced apart from each other in the first (X) direction, as described in FIGS. **13** and **14**, may be replaced with a lower portion **189**La″ formed in a linear shape entirely extending in the first (X) direction.
(113) Next, a modified example of the separation structures **189** in plan view, described with reference to FIGS. **9**A to **9**E, will be described with reference to FIGS. **16**A to **16**C. FIG. **16**A may be a plan view corresponding to FIG. **9**A and illustrating modified components of the components of FIG. **9**A, FIG. **16**B may be a plan view corresponding to FIG. **9**C and illustrating modified components of the components of FIG. **9**C, and FIG. **16**C may be a plan view corresponding to FIG. **9**E and illustrating modified components of the components of FIG. **9**E.
(114) Referring to FIGS. **16**A to **16**C, the second portion **189**U**1_2** of the wave pattern, as described in FIG. **9**A, may be replaced with a second portion (**189**U**1_2**′ in FIG. **16**A) having a side surface in which straight portions LP and concave portions CP are repeatedly arranged. The plurality of third portions **189**U**1_3** respectively having a circular shape, as described in FIG. **9**C, may be replaced with a plurality of third portions **189**U**1_3**′ respectively having a bar shape. The plurality of sixth portions **189**U**2_3** respectively having a circular shape, as described in FIG. **9**E, may be replaced with a plurality of sixth portions **189**U**2_3**′ respectively having a bar shape.
(115) Next, a modified example of a semiconductor device according to an example embodiment of the present inventive concepts will be described with reference to FIGS. **17**A and **17**B. FIG. **17**A may be a cross-sectional view schematically illustrating a region taken along line Ib-Ib′ and a region taken along line IIb-IIb′ in FIGS. **9**A to **9**E, and FIG. **17**B may be a cross-sectional view schematically illustrating a region taken along line IIIb-IIIb′ and a region taken along line IVb-IVb′ in FIGS. **9**A to **9**E.
(116) Referring to FIGS. **17**A and **17**B, in a semiconductor device **1***b*, the upper structure **59**, as described in FIGS. **12**A and **12**B, may be replaced with an upper structure **159**, which further includes an intermediate stack structure ST_M and an intermediate capping insulating layer **136** covering at least one side of the intermediate stack structure ST_M. The intermediate stack structure ST_M may be disposed between the lower stack structure ST_L and the upper stack structure ST_U, as described with reference to FIGS. **12**A and **12**B. The intermediate stack structure ST_M may include intermediate interlayer insulating layers **130** and intermediate gate layers **133***g*, alternately stacked. In the intermediate interlayer insulating layers **130**, a lowermost intermediate interlayer insulating layer **130**L may be disposed on an uppermost lower interlayer insulating layer **30**U of a lower interlayer insulating layers **30**, and an uppermost intermediate interlayer insulating layer **130**U may be disposed below a lowermost upper interlayer insulating layer **45**L among upper interlayer insulating layers **45**.
(117) A stack structure ST′ including the lower stack structure ST_L, the intermediate stack structure ST_M, and the upper stack structure ST_U, sequentially stacked, may be configured. Therefore, the stack structure ST′ may include interlayer insulating layers and gate layers, alternately stacked.
(118) The capping insulating structure **153** may include the intermediate capping insulating layer **136**, together with the lower and upper capping insulating layers **36** and **51**, as described in FIGS. **12**A and **12**B.
(119) In the semiconductor device **1***b*, the vertical memory structures **65**, as described in FIGS. **12**A and **12**B, may be replaced with vertical memory structures **165** including a portion penetrating

through the intermediate stack structure ST_M. The vertical support structures **75**, as described in FIGS. **12**A and **12**B, may be replaced with vertical support structures **175** including a portion penetrating through the intermediate stack structure ST_M.

(120) For example, a side surface of each of the vertical memory structures **165** may include a plurality of slope change portions **165**V**1** and **165**V**2** corresponding to the slope change portion (**65**V of FIG. **3**) described with reference to FIG. **3**. For example, a side surface of each of the vertical memory structures **165** may include a first slope change portion **165**V**1** and a second slope change portion **165**V**2** having a level higher than that of the first slope change portion **165**V**1**.

(121) The separation structures **189**, as described in FIGS. **12**A and **12**B, may be replaced with separation structures **289** including a portion penetrating through the intermediate stack structure ST_M. The separation structures **289** may include first separation structures **289_1** parallel to each other, and a second separation structure **289_2** and a third separation structure **289_3** arranged between the first separation structures **289_1** and having opposite end portions (e.g., being arranged in a row). At least in a portion penetrating through the stack structure ST′, each of the separation structures **289** may include a plurality of line portions **289**LN**1**, **289**LN**2**, and **289**LN**3** spaced apart from each other in the vertical (Z) direction and overlapping each other, and pillar portions **289**P**1**, **289**P**2**, and **289**P**3** disposed above the plurality of line portions **289**LN**1**, **289**LN**2**, and **289**LN**3**, respectively. For example, in each of the separation structures **289**, the plurality of line portions **289**LN**1**, **289**LN**2**, and **289**LN**3** may include a first line portion **289**LN**1**, a second line portion **289**LN**2** on the first line portion **289**LN**1**, and a third line portion **289**LN**3** on the second line portion **289**LN**2**, wherein the pillar portions **289**P**1**, **289**P**2**, and **289**P**3** may include a plurality of first pillar portions **289**P**1** spaced apart from each other in the first (X) direction between the first line portion **289**LN**1** and the second line portion **289**LN**2**, a plurality of second pillar portions **289**P**2** spaced apart from each other in the first (X) direction between the second line portion **289**LN**2** and the third line portion **289**LN**3**, and a plurality of third pillar portions **289**P**3** spaced apart from each other in the first (X) direction on the third line portion **289**LN**3**.

(122) The semiconductor device **1***b* may include a first intermediate support structure SP**1** disposed between the plurality of first pillar portions **289**P**1**, a second intermediate support structure SP**3** disposed between the plurality of second pillar portions **289**P**2**, and an upper support structure SP**2** disposed between the plurality of third pillar portions **289**P**3**.

(123) The first intermediate support structure SP**1** may be formed of at least a portion of the uppermost lower interlayer insulating layer **30**U and at least a portion of the lowermost intermediate interlayer insulating layer **130**L. The second intermediate support structure SP**3** may be formed of at least a portion of the uppermost intermediate interlayer insulating layer **130**U and at least a portion of the lowermost upper interlayer insulating layer **45**L. The upper support structure SP**2** may be formed of at least a portion of the uppermost upper interlayer insulating layer **45**U.

(124) In each of the separation structures **289**, the lower portion **289**L may have the same or substantially similar structure as the lower portion of any one separation structure of the above-described example embodiments. For example, in each of the separation structures **289**, the lower portion **289**L may have the same or substantially similar structure as the lower portion **189**L of the separation structures **189**, as described with reference to FIGS. **10**A and **10**B.

(125) Next, an example of a method of forming a semiconductor device according to an example embodiment of the present inventive concepts will be described with reference to FIGS. **18** to **25**F. FIGS. **18** to **25**F are views illustrating an example of a method of forming a semiconductor device according to an example embodiment of the present inventive concepts.

(126) In FIGS. **18** to **25**F, FIG. **18** may be a plan view schematically illustrating an example of a method of forming a semiconductor device according to an example embodiment of the present inventive concepts, FIGS. **19**A, **20**A, **21**A, **22**A, and **23**A may be a cross-sectional view schematically illustrating a region taken along line I-I′ and a region taken along line II-II′ of FIG.

18, FIGS. **22**B and **23**B may be a cross-sectional view schematically illustrating a region taken along line III-III′ and a region taken along line IV-IV′ of FIG. **18**A, FIG. **24** may be a partial enlarged view schematically illustrating portion 'C' in FIG. **23**A, and FIGS. **25**A to **25**F may be partial enlarged views corresponding to the partially enlarged view of FIG. **24**.
(127) Referring to FIGS. **18**, **19**A, and **19**B, a peripheral circuit structure **3** may be formed. The peripheral circuit structure **3** may include a semiconductor substrate **5**, an device isolation layer **7***s* defining an active region **7***a* in the semiconductor substrate **5**, a peripheral circuit **9** on the semiconductor substrate **5**, a circuit interconnection **11** electrically connected to the peripheral circuit **9** on the semiconductor substrate **5**, and a lower insulating layer **13** covering the peripheral circuit **9** and the circuit interconnection **11** on the semiconductor substrate **5**. The peripheral circuit **9** may include a peripheral gate **9***a* on the active region **7***a*, and a peripheral source/drain **9***b* disposed in the active region **7***a* on both sides of the peripheral gate **9***a*.
(128) A plate layer **15** may be formed on the peripheral circuit structure **3**. The plate layer **15** may be formed as a silicon layer, for example, a doped silicon layer. For example, the plate layer **15** may be formed as a silicon layer having N-type conductivity.
(129) Trenches **17** may be formed in the plate layer **15**. A first intermediate layer **21***a* and a second intermediate layer **21***b*, covering inner walls of the trenches **17**, extending onto an upper surface of the plate layer **15**, and sequentially stacked, may be formed. The first intermediate layer **21***a* may be formed of silicon oxide, and the second intermediate layer **21***b* may be formed of silicon nitride. A thickness of the second intermediate layer **21***b* may be greater than a thickness of the first intermediate layer **21***a*.
(130) A sacrificial layer may be formed on the second intermediate layer **21***b*, and the sacrificial layer may be planarized to form sacrificial gap-fill layers **23**, filling the trenches **17**, on the second intermediate layer **21***b*.
(131) The sacrificial gap-fill layers **23** may be formed of a silicon material, but example embodiments are not limited thereto, and other materials may be substituted.
(132) A third intermediate layer **21***c* may be formed on the second intermediate layer **21***b* and the sacrificial gap-fill layers **23**. The third intermediate layer **21***c* may be formed of silicon oxide.
(133) The first to third intermediate layers **21***a*, **21***b*, and **21***c* may be referred to as an intermediate layer **21**.
(134) An intermediate insulating layer **25** penetrating through the intermediate layer **21** and the plate layer **15** may be formed. The intermediate insulating layer **25** may be formed of silicon oxide.
(135) An upper pattern layer **27** may be formed on the intermediate layer **21** and the intermediate insulating layer **25**. The upper pattern layer **27** may be formed of a silicon material.
(136) In an example, the intermediate layer **21** may have an opening exposing the plate layer **15**, and the upper pattern layer **27** may have a portion to penetrate through the opening of the intermediate layer **21** to contact the plate layer **15**.
(137) Referring to FIGS. **18**, **20**A, and **20**B, first lower mold structures **30***a* and **33***a* may be formed on the upper pattern layer **27**. The first lower mold structures **30***a* and **33***a* may include first lower interlayer insulating layers **30***a* and first lower sacrificial gate layers **33***a*, alternately stacked.
(138) At least one side of the first lower mold structures **30***a* and **33***a* may have a stepped shape.
(139) Before forming the first lower mold structures **30***a* and **33***a* or while forming the first lower mold structures **30***a* and **33***a*, the upper pattern layer **27** may be patterned. A side surface of the upper pattern layer **27** may not be aligned with a side surface of the plate layer **15**.
(140) Referring to FIGS. **18**, **21**A, and **21**B, a second lower mold structure may be formed on the first lower mold structures **30***a* and **33***a*. The second lower mold structure may include second lower interlayer insulating layers and second lower sacrificial gate layers, alternately stacked.
(141) In some example embodiments, the first lower interlayer insulating layers **30***a* and the second lower interlayer insulating layers may be referred to as lower interlayer insulating layers **30**, and the first lower sacrificial gate layers **33***a* and the second lower sacrificial gate layers will be

referred to as lower sacrificial gate layers **33**.

(142) The lower interlayer insulating layers **30** and the lower sacrificial gate layers **33** may form lower mold structures **30** and **33**. A lower capping insulating layer **36** covering at least one side of the lower mold structures **30** and **33** may be formed. Among the lower interlayer insulating layers **30** and the lower sacrificial gate layers **33**, an uppermost layer may be an uppermost lower interlayer insulating layer **30**U.

(143) The lower capping insulating layer **36** and the lower mold structures **30** and **33** may constitute a first preliminary upper structure.

(144) Lower sacrificial structures **42***a*, **42***b*, and **42***c* penetrating through the first preliminary upper structures **30**, **33**, and **36** and extending in a downward direction may be formed. The lower sacrificial structures **42***a*, **42***b*, and **42***c* may include lower sacrificial separation structures **42***a*, lower sacrificial vertical memory structures **42***b*, and lower sacrificial vertical support structures **42***c*.

(145) The lower sacrificial separation structures **42***a* may sequentially penetrate through the first preliminary upper structures **30**, **33**, and **36** and the upper pattern layer **27**, and may extend into the sacrificial gap-fill layers **23**.

(146) The sacrificial gap-fill layers **23** may serve as an etch stop layer. For example, in an etching process for forming the lower sacrificial separation structures **42***a*, since the sacrificial gap-fill layers **23** may serve as an etch stop layer, the lower sacrificial separation structures **42***a* may not be in contact with the plate layer **15**. Therefore, it is possible to prevent the occurrence of defects while simultaneously forming the lower sacrificial structures **42***a*, **42***b*, and **42***c*.

(147) The lower sacrificial vertical memory structures **42***b* and the lower sacrificial vertical support structures **42***c* may sequentially penetrate through the first preliminary upper structure **30**, **33**, and **36**, the upper pattern layer **27**, and the intermediate layer **21**, and, may extend into the plate layer **15**.

(148) Referring to FIGS. **18**, **22**A and **22**B, upper mold structures **45** and **48** may be formed on the first preliminary upper structures **30**, **33**, and **36**. The upper mold structures **45** and **48** may include upper interlayer insulating layers **45** and upper sacrificial gate layers **48**, alternately stacked. Among the upper interlayer insulating layers **45** and the upper sacrificial gate layers **48**, an uppermost portion may be an uppermost upper interlayer insulating layer **45**U. At least one side of the upper mold structures **45** and **48** may have a stepped shape. An upper capping insulating layer **51** covering at least one side of the upper mold structures **45** and **48** and covering the lower capping insulating layer **36** may be formed.

(149) The lower capping insulating layer **36** and the upper capping insulating layer **51** may constitute a capping insulating structure **53**.

(150) The upper mold structures **45** and **48** and the upper capping insulating layer **51** may constitute second preliminary upper structures **45**, **48**, and **51**.

(151) Upper sacrificial structures penetrating through the second preliminary upper structures **45**, **48**, and **51** and contacting the lower sacrificial structures **42***a*, **42***b*, and **42***c* may be simultaneously formed. For example, the upper sacrificial structures may include upper sacrificial separation structures **63** contacting the lower sacrificial separation structures **42***a*.

(152) After forming the upper sacrificial structures, a portion of the upper sacrificial structures contacting the lower sacrificial vertical memory structures **42***b* and the lower sacrificial support structures **42***c* may be removed, the lower sacrificial vertical memory structures **42***b* and the lower sacrificial support structures **42***c* may be removed to form holes, and vertical memory structures **65** and vertical support structures **75** may be formed in the holes.

(153) In an example, the vertical memory structures **65** and the vertical support structures **75** may be formed simultaneously and may include layers having the same material.

(154) In another example, after forming the vertical memory structures **65**, the vertical support structures **75** may be formed.

(155) Subsequently, a mask **77** covering the vertical memory structures **65** and the vertical support structures **75** and exposing the upper sacrificial separation structures **63** may be formed.
(156) Referring to FIGS. **18**, **23**A, **23**B, and **24**, the upper sacrificial separation structures **63** exposed from the mask **77** may be removed, and the lower sacrificial separation structures **42***a* exposed by the removal of the upper sacrificial separation structures **63** may be removed, to form separation holes **79** exposing the sacrificial gap-fill layers **23**.
(157) Next, a subsequent process will be described with reference to FIGS. **25**A to **25**F. In this case, any one separation hole of the separation holes **79** will be mainly described.
(158) Referring to FIG. **25**A, the lower and upper interlayer insulating layers **30** and **45** and the third intermediate layer **21***c*, exposed by the separation holes **79**, may be partially etched to form a partially expanded separation hole **79***a*.
(159) Referring to FIG. **25**B, the upper pattern layer **27** and the sacrificial gap-fill layer **23** may be partially etched to form a partially expanded separation hole **79***b*.
(160) Referring to FIG. **25**C, the upper pattern layer **27** and the sacrificial gap-fill layer **23**, partially etched, may be oxidized to form a sacrificial oxide layer **81**.
(161) Referring to FIG. **25**D, the lower and upper sacrificial gate layers **33** and **48** may be partially etched, to form the separation hole **79***b* as a separation opening **79***c* having a linear shape. For example, separation holes **79***b* adjacent to each other may be connected to each other to form a separation opening **79***c*.
(162) Referring to FIG. **25**E, a sacrificial spacer **86** may be formed on a side surface of the separation opening **79***c*.
(163) Referring to FIG. **25**F, the second intermediate layer **21***b* may be etched using the sacrificial spacer **86** as an etching mask, and the first and third intermediate layers **21***a* and **21***c* and the data storage structure **67** may be etched to form an empty space. A first pattern layer **19***a* may be formed in the empty space. While the first and third intermediate layers **21***a* and **21***c* and the data storage structure **67** are etched, the sacrificial spacer **86** may be etched and removed.
(164) Again, referring to FIGS. **1**, **2**A, and **2**B, the lower and upper sacrificial gate layers **33** and **48** may be selectively removed to form empty spaces, and gate layers may be formed in the empty spaces.
(165) Subsequently, separation structures **89** filling the separation openings **79***d* may be formed.
(166) Next, a modified example of a semiconductor device according to an example embodiment of the present inventive concepts will be described with reference to FIG. **26**.
(167) FIG. **26** may be a cross-sectional view schematically illustrating a region taken along line I-I′ and a region taken along line II-II′ of FIG. **1**.
(168) Referring to FIG. **26**, a semiconductor device Ic may include a first structure **103** and a second structure **3**′ on the first structure **103**.
(169) The first structure **103** may be a memory chip, and the second structure **3**′ may be a logic chip. The second structure **3**′ may be a peripheral circuit structure.
(170) The first structure **103** may include a plate layer **15**, an upper structure **59**, vertical memory structures **65**, vertical support structures **75**, and separation structures **89**, as described with reference to FIGS. **1** to **2**B.
(171) The first structure **103** may further include bit line contact plugs **325** disposed on the upper structure **59** and electrically connected to the vertical memory structures **65**, and bit lines **327** on the bit line contact plugs **325**.
(172) The first structure **103** may further include an insulating layer **320** disposed on the upper structure **59** and covering the bit line contact plugs **325** and the bit lines **327**. The first structure **103** may further include first bonding metal patterns **315** coplanar with an upper surface of the insulating layer **320** and buried in the insulating layer **320**.
(173) The second structure **3**′ may include a semiconductor substrate **5**, a peripheral circuit **9** including a peripheral gate **9***a* and a peripheral source/drain **9***b* on the semiconductor substrate **5**, a

circuit interconnection **11** electrically connected to the peripheral circuit **9** on the semiconductor substrate **5**, and an insulating layer **13** covering the peripheral circuit **9** and the circuit interconnection **11** on the semiconductor substrate **5**. The second structure **3**′ may further include second bonding metal patterns **215** coplanar with an upper surface of the insulating layer **13** and buried in the insulating layer **13**. The upper surface of the insulating layer **320** and upper surfaces of the first bonding metal patterns **315** in the first structure **103** may be in contact with and connected to the upper surface of the insulating layer **13** and upper surfaces of the second bonding metal patterns **215** in the second structure **3**′, respectively.
(174) Next, a modified example of the semiconductor device according to an example embodiment of the present inventive concepts will be described with reference to FIG. **27**.
(175) FIG. **27** may be a cross-sectional view schematically illustrating a region taken along line Ib-Ib′ and a region taken along line IIb-IIb′ of FIGS. **9**A to **9**E.
(176) Referring to FIG. **27**, a semiconductor device **1***d* may include a first structure **103**′, and a second structure **3**′ disposed on the first structure **103**′ and substantially identical to that described with reference to FIG. **26**.
(177) The first structure **103**′ may include a plate layer **15**, an upper structure **59**, vertical memory structures **65**, vertical support structures **75**, and separation structures **189**, as described with reference to FIGS. **9**A to **11**. The first structure **103**′ may further include bit line contact plugs **325** disposed on the upper structure **59** and electrically connected to the vertical memory structures **65**, and bit lines **327** on the bit line contact plugs **325**. The first structure **103**′ may further include an insulating layer **320** disposed on the upper structure **59** and covering the bit line contact plugs **325** and the bit lines **327**. The first structure **103**′ may further include first bonding metal patterns **315** coplanar with an upper surface of the insulating layer **320** and buried in the insulating layer **320**. The upper surface of the insulating layer **320** and upper surfaces of the first bonding metal patterns **315** in the first structure **103**′ may be in contact with and connected to an upper surface of the insulating layer **13** and upper surfaces of second bonding metal patterns **215** in the second structure **3**′, respectively.
(178) Next, a data storage system including a semiconductor device according to an example embodiment of the present inventive concepts will be described with reference to FIGS. **28** and **29**, respectively. FIG. **28** is a view schematically illustrating a data storage system including a semiconductor device according to an example embodiment of the present inventive concepts, and FIG. **29** is a perspective view schematically illustrating a data storage system including a semiconductor device according to an example embodiment of the present inventive concepts.
(179) Referring to FIG. **28**, a data storage system **1000** may include a semiconductor device **1100**, and a controller **1200** electrically connected to the semiconductor device **1100** and controlling the semiconductor device **1100**. The data storage system **1000** may be a storage device including the semiconductor device **1100**, or an electronic device including the storage device. For example, the data storage system **1000** may be a solid state drive device (SSD), a universal serial bus (USB), a computing system, a medical device, or a communication device, including the semiconductor device **1100**.
(180) In an example embodiment, the data storage system **1000** may be an electronic system storing data.
(181) The semiconductor device **1100** may be a semiconductor device according to any one of the example embodiments described above with reference to FIGS. **1** to **27**. The semiconductor device **1100** may include a first structure **1100**F, and a second structure **1100**S on the first structure **1100**F.
(182) The first structure **1110**F may include the peripheral circuit structure **3** described above, and the second structure **1100**S may include the plate layer **15**, the pattern structure **19**, the upper pattern layer **27**, the upper structure **59** or **159**, the vertical memory structures **65**, the vertical support structures **75** and the separation structures **89** or **189**, described above.
(183) The first structure **1100**F may be a peripheral circuit structure including a decoder circuit

1110, a page buffer **1120**, and a logic circuit **1130**. For example, the first structure **1100**F may include the peripheral circuit **9** described above. The peripheral circuit **9** may be a transistor constituting the peripheral circuit structure including the decoder circuit **1110**, the page buffer **1120**, and the logic circuit **1130**.
(184) The second structure **1100**S may be a memory cell structure including bit lines BL, a common source line CSL, word lines WL, first and second upper gate lines UL**1** and UL**2**, first and second lower gate lines LL**1** and LL**2**, and memory cell strings CSTR between each of the bit lines BL and the common source line CSL.
(185) The plate layer **15** and the first pattern layer **19***a*, described above, may include a silicon layer having N-type conductivity, and the silicon layer having N-type conductivity may be the common source line CSL.
(186) In the second structure **1100**S, each of the memory cell strings CSTR may include lower transistors LT**1** and LT**2** adjacent to the common source line CSL, upper transistors UT**1** and UT**2** adjacent to each of the bit lines BL, and a plurality of memory cell transistors MCT disposed between each of the lower transistors LT**1** and LT**2** and each of the upper transistors UT**1** and UT**2**. The number of lower transistors LT**1** and LT**2** and the number of upper transistors UT**1** and UT**2** may be variously changed according to example embodiments.
(187) In some example embodiments, each of the upper transistors UT**1** and UT**2** may include a string select transistor, and each of the lower transistors LT**1** and LT**2** may include a ground select transistor. The lower gate lines LL**1** and LL**2** may be gate electrodes of the lower transistors LT**1** and LT**2**, respectively. The word lines WL may be gate electrodes of the memory cell transistors MCT, and the upper gate lines UL**1** and UL**2** may be gate electrodes of the upper transistors UT**1** and UT**2**, respectively.
(188) The lower and upper gate layers **33***g* and **48***g*, described above, may constitute the lower gate lines LL**1** and LL**2**, the word lines WL, and the upper gate lines UL**1** and UL**2**.
(189) In some example embodiments, the lower transistors LT**1** and LT**2** may include a lower erase control transistor LT**1** and a ground select transistor LT**2**, connected in series. The upper transistors UT**1** and UT**2** may include a string select transistor UT**1** and an upper erase control transistor UT**2**, connected in series. At least one of the lower erase control transistor LT**1** or the upper erase control transistor UT**2** may be used for an erase operation of erasing data stored in the memory cell transistors MCT using a gate-induced-drain-leakage (GIDL) phenomenon.
(190) The common source line CSL, the first and second lower gate lines LL**1** and LL**2**, the word lines WL, and the first and second upper gate lines UL**1** and UL**2** may be electrically connected to the decoder circuit **1110** through first connection wirings **1115** extending from the first structure **1100**F into the second structure **1100**S.
(191) The bit lines BL may be electrically connected to the page buffer **1120** through second connection wirings **1125** extending from the first structure **1100**F into the second structure **1100**S. The bit lines BL may be the bit lines (**327** of FIGS. **26** and **27**) described above.
(192) In the first structure **1100**F, the decoder circuit **1110** and the page buffer **1120** may perform a control operation on at least one selected memory cell transistor among the plurality of memory cell transistors MCT. The decoder circuit **1110** and the page buffer **1120** may be controlled by the logic circuit **1130**.
(193) The semiconductor device **1100** may further include an input/output pad **1101**. The semiconductor device **1100** may communicate with the controller **1200** through the input/output pad **1101** electrically connected to the logic circuit **1130**. The input/output pad **1101** may be electrically connected to the logic circuit **1130** through input/output connection wirings **1135** extending from the first structure **1100**F into the second structure **1100**S. Therefore, the controller **1200** may be electrically connected to the semiconductor device **1100** through the input/output pad **1101**, and may control the semiconductor device **1100**.
(194) The controller **1200** may include a processor **1210**, a NAND controller **1220**, and a host

interface **1230**. According to some example embodiments, the data storage system **1000** may include a plurality of semiconductor devices **1100**, and in this case, the controller **1200** may control the plurality of semiconductor devices **1100**.
(195) The processor **1210** may control an overall operation of the data storage system **1000** including the controller **1200**. The processor **1210** may operate according to a predetermined firmware, and may access to the semiconductor device **1100** by controlling the NAND controller **1220**. The NAND controller **1220** may include a NAND interface **1221** processing communications with the semiconductor device **1100**. A control command for controlling the semiconductor device **1100**, data to be written to the memory cell transistors MCT of the semiconductor device **1100**, data to be read from the memory cell transistors MCT of the semiconductor device **1100**, or the like may be transmitted through the NAND interface **1221**. The host interface **1230** may provide a communication function between the data storage system **1000** and an external host. When a control command is received from the external host through the host interface **1230**, the processor **1210** may control the semiconductor device **1100** in response to the control command.
(196) FIG. **29** is a perspective view schematically illustrating a data storage system including a semiconductor device according to an example embodiment.
(197) Referring to FIG. **29**, a data storage system **2000** according to an example embodiment of the present inventive concepts may include a main substrate **2001**, a controller **2002** mounted on the main substrate **2001**, at least one semiconductor package **2003**, and a DRAM **2004**. The semiconductor package **2003** and the DRAM **2004** may be connected to the controller **2002** by wiring patterns **2005** formed on the main substrate **2001**.
(198) The main substrate **2001** may include a connector **2006** including a plurality of pins, which may be coupled to an external host. The number and an arrangement of the plurality of pins in the connector **2006** may vary according to a communication interface between the data storage system **2000** and the external host. In some example embodiments, the data storage system **2000** may be communicated with the external host according to any one interface of a universal serial bus (USB), peripheral component interconnection express (PCI-Express), serial advanced technology attachment (SATA), M-Phy for universal flash storage (UFS), or the like. In some example embodiments, the data storage system **2000** may be operated by power supplied from the external host through the connector **2006**. The data storage system **2000** may further include a power management integrated circuit (PMIC) distributing power, supplied from the external host, to the controller **2002** and the semiconductor package **2003**.
(199) The controller **2002** may write data to the semiconductor package **2003** or read data from the semiconductor package **2003**, and may improve an operation speed of the data storage system **2000**.
(200) The DRAM **2004** may be a buffer memory reducing a difference in speed between the semiconductor package **2003**, which may be a data storage space, and the external host. The DRAM **2004** included in the data storage system **2000** may also operate as a type of cache memory, and may provide a space temporarily storing data in a control operation on the semiconductor package **2003**. When the DRAM **2004** is included in the data storage system **2000**, the controller **2002** may further include a DRAM controller controlling the DRAM **2004** in addition to a NAND controller controlling the semiconductor package **2003**.
(201) The semiconductor package **2003** may include first and second semiconductor packages **2003***a* and **2003***b*, spaced apart from each other. Each of the first and second semiconductor packages **2003***a* and **2003***b* may be a semiconductor package including a plurality of semiconductor chips **2200**. Each of the semiconductor chips **2200** may include the semiconductor device according to any one of the example embodiments described above with reference to FIGS. **1** to **27**. For example, each of the semiconductor chips **2200** may include vertical memory structures **3220**, which may correspond to the vertical memory structures **65** described above, and

separation structures **3210**, which may correspond to the separation structures **89**, **189**, or **289**, described above, respectively.
(202) Each of the first and second semiconductor packages **2003***a* and **2003***b* may include a package substrate **2100**, semiconductor chips **2200** on the package substrate **2100**, adhesive layers **2300** disposed on a lower surface of each of the semiconductor chips **2200**, a connection structure **2400** electrically connecting each of the semiconductor chips **2200** and the package substrate **2100**, and a molding layer **2500** covering the semiconductor chips **2200** and the connection structure **2400** on the package substrate **2100**.
(203) The package substrate **2100** may be a printed circuit board including package upper pads **2130**. Each of the semiconductor chips **2200** may include an input/output pad **2210**.
(204) In some example embodiments, the connection structure **2400** may be a bonding wire electrically connecting the input/output pad **2210** and the upper package pads **2130**. Therefore, in each of the first and second semiconductor packages **2003***a* and **2003***b*, the semiconductor chips **2200** may be electrically connected to each other by a bonding wire process, and may be electrically connected to the package upper pads **2130** of the package substrate **2100**. According to some example embodiments, in each of the first and second semiconductor packages **2003***a* and **2003***b*, the semiconductor chips **2200** may be electrically connected to each other by a connection structure including a through silicon via (TSV), instead of a connection structure **2400** by a bonding wire process.
(205) In some example embodiments, the controller **2002** and the semiconductor chips **2200** may be included in one (1) package. In an example embodiment, the controller **2002** and the semiconductor chips **2200** may be mounted on a separate interposer substrate, different from the main substrate **2001**, and the controller **2002** and the semiconductor chips **2200** may be connected to each other by a wiring formed on the interposer substrate.
(206) According to some example embodiments of the inventive concepts, a semiconductor device including a separation structure including a lower portion extending into a plate layer and an upper portion having a width smaller than the maximum width of the lower portion may be provided. A structure of the separation structure may improve reliability for the semiconductor device.
(207) According to some example embodiments of the inventive concepts, a semiconductor device including a separation structure including a line portion and pillar portions overlapping the line portion may be provided. A structure of the separation structure may mitigate or prevent a stack structure from being collapsed or deformed. Therefore, reliability of the semiconductor device may be improved.
(208) Various advantages and effects of the present inventive concepts are not limited to the above, and will be more easily understood in the process of describing specific example embodiments of the present inventive concepts.
(209) While some example embodiments have been illustrated and described above, it will be apparent to those skilled in the art that modifications and variations could be made without departing from the scope of the present inventive concepts as defined by the appended claims.

Claims

1. A semiconductor device comprising: a plate layer; a pattern structure on the plate layer; an upper pattern layer on the pattern structure; an upper structure including a stack structure and a capping insulating structure, the stack structure being on the upper pattern layer and including interlayer insulating layers and gate layers alternately stacked on each other, the capping insulating structure covering at least a portion of the stack structure; separation structures penetrating through the upper structure, the upper pattern layer, and the pattern structure, and extending into the plate layer; and vertical memory structures penetrating through the upper structure, the upper pattern layer, and the pattern structure between the separation structures, extending into the plate layer, and contacting

the plate layer, wherein each of the separation structures includes a lower portion and an upper portion on the lower portion, the lower portion includes a first lower portion, a second lower portion, and a third lower portion, the first lower portion is in the plate layer and contacts the plate layer, the second lower portion penetrates through the pattern structure, the third lower portion penetrates through the upper pattern layer, the upper portion penetrates through the upper structure, and a maximum width of the first lower portion is different from a maximum width of the third lower portion.

2. The semiconductor device of claim 1, wherein the stack structure includes a lower stack structure and an upper stack structure on the lower stack structure, the lower stack structure includes lower interlayer insulating layers and lower gate layers alternately stacked on each other, the upper stack structure includes upper interlayer insulating layers and upper gate layers stacked alternately on each other, and a side surface of each of the vertical memory structures includes a slope change portion in which a side slope changes, at a height level between a lowermost upper gate layer among the upper gate layers and an uppermost lower gate layer among the lower gate layers.

3. The semiconductor device of claim 2, wherein a side surface of each of the separation structures includes a slope change portion, in which a side slope changes, at a height level between the lowermost upper gate layer among the upper gate layers and the uppermost lower gate layer among the lower gate layers.

4. The semiconductor device of claim 1, wherein in plan view, in each of the separation structures, a side shape of the first lower portion is different from a side shape of the upper portion.

5. The semiconductor device of claim 1, wherein in plan view, in each of the separation structures, a side surface of the first lower portion has a straight linear shape, and a side surface of the upper portion has a wavy shape.

6. The semiconductor device of claim 1, wherein in plan view, in each of the separation structures, a side surface of the first lower portion has a straight linear shape, and a side surface of the third lower portion has a wavy shape.

7. The semiconductor device of claim 1, wherein the separation structures include first separation structures parallel to each other, and a second separation structure and a third separation structure between the first separation structures, an upper portion of the second separation structure and an upper portion of the third separation structure are spaced apart from each other, the separation structures further include a connection portion connecting a lower portion of the second separation structure and a lower portion of the third separation structure, and the stack structure includes a portion between the upper portion of the second separation structure and the upper portion of the third separation structure.

8. The semiconductor device of claim 1, wherein the separation structures include first separation structures parallel to each other, and a second separation structure and a third separation structure between the first separation structures and being in a row, and a portion of the plate layer is between the second separation structure and the third separation structure.

9. The semiconductor device of claim 1, further comprising: a peripheral circuit structure below the plate layer, wherein at least a portion of the plate layer is between the peripheral circuit structure and the upper structure, and the peripheral circuit structure includes a semiconductor substrate and a peripheral circuit on the semiconductor substrate.

10. The semiconductor device of claim 1, further comprising: a peripheral circuit structure on the upper structure, wherein the peripheral circuit structure includes a semiconductor substrate and a peripheral circuit on the semiconductor substrate, and the peripheral circuit is between the semiconductor substrate and the upper structure.

11. The semiconductor device of claim 1, wherein at least one of the separation structures includes a line portion extending in a first direction, and pillar portions connected to the line portion at a different height level from the line portion, the pillar portions being spaced apart from each other in

the first direction.

12. The semiconductor device of claim 1, wherein the stack structure includes a lower stack structure and an upper stack structure on the lower stack structure, the lower stack structure includes lower interlayer insulating layers and lower gate layers alternately stacked on each other, the upper stack structure includes upper interlayer insulating layers and upper gate layers alternately stacked on each other, and a first separation structure among the separation structures includes a lower line portion passing through the lower gate layers, an upper line portion passing through the upper gate layers, and an intermediate pillar portion spaced apart from each other between the lower line portion and the upper line portion.

13. The semiconductor device of claim 12, wherein the first separation structure of the separation structures further include upper pillar portions spaced apart from each other on the upper line portion.

14. A semiconductor device comprising: a plate layer; an upper structure including a stack structure and a capping insulating structure covering at least a portion of the stack structure, the stack structure on the plate layer and including interlayer insulating layers and gate layers alternately stacked on each other; separation structures penetrating through at least a portion of the upper structure; and vertical memory structures penetrating through the upper structure between the separation structures, wherein a first separation structure among the separation structures includes a first line portion extending in a first direction and being parallel to an upper surface of the plate layer, and a plurality of first pillar portions being at a different height level from the first line portion and spaced apart from each other in the first direction, and the plurality of first pillar portions are at a level lower than that of an uppermost gate layer among the gate layers, and at a level higher than that of a lowermost gate layer among the gate layers.

15. The semiconductor device of claim 14, further comprising: a support structure overlapping the first line portion and being between the plurality of first pillar portions, wherein at least a portion of the support structure is at a same level as at least a portion of an interlayer insulating layer among the interlayer insulating layers.

16. The semiconductor device of claim 14, wherein the stack structure includes a lower stack structure and an upper stack structure on the lower stack structure, the lower stack structure includes lower interlayer insulating layers and lower gate layers alternately stacked on each other, the upper stack structure includes upper interlayer insulating layers and upper gate layers alternately stacked on each other, the first line portion penetrates through the lower gate layers, and the plurality of first pillar portions extend from the first line portion in a vertical direction perpendicular to an upper surface of the plate layer.

17. The semiconductor device of claim 16, wherein the first separation structure further includes a second line portion penetrating through the upper gate layers and extending in the first direction, and upper pillar portions being on the second line portion and spaced apart from each other in the first direction.

18. A data storage system comprising: a semiconductor device including an input/output pad; and a controller electrically connected to the semiconductor device through the input/output pad and configured to control the semiconductor device, the semiconductor device including, a plate layer, a pattern structure on the plate layer, an upper pattern layer on the pattern structure, an upper structure including a stack structure and a capping insulating structure covering at least a portion of the stack structure, the stack structure on the plate layer and including interlayer insulating layers and gate layers alternately stacked on each other, separation structures penetrating through the upper structure, the upper pattern layer, and the pattern structure, and extending into the plate layer, and vertical memory structures penetrating through the upper structure, the upper pattern layer, and the pattern structure between the separation structures, extending into the plate layer, and contacting the plate layer, wherein each of the separation structures includes a lower portion and an upper portion on the lower portion, the lower portion includes a first lower portion, a second lower

portion, and a third lower portion, the first lower portion is in the plate layer and contacts the plate layer, the second lower portion penetrates through the pattern structure, the third lower portion penetrates through the upper pattern layer, the upper portion penetrates through the upper structure, and a maximum width of the first lower portion is different from a maximum width of the third lower portion.

19. The data storage system of claim 18, wherein, in plan view, in each of the separation structures, a side surface of the first lower portion has a straight linear shape, and a side surface of the upper portion has a wavy shape.

20. The data storage system of claim 18, wherein at least one of the separation structures includes a line portion extending in a first direction, and pillar portions connected to the line portion at a different height level from the line portion, the pillar portions being spaced apart from each other in the first direction.